Round-Robin Game Turns

Category B Coding Assignment - Problem 5

Problem summary

Store players in a circular singly linked list with a tail reference. Head is tail.next; traversal wraps back to the head automatically.

Java solution

```
public class Main {
 static class Node {String name;Node next;Node(String n){name=n;}}
 static class CircularPlayers {Node tail;
  void addLast(String name){Node n=new Node(name);if(tail==null){tail=n;n.next=n;}else{n.next=tail.next;tail.nex
  void printTurns(int turns){if(tail==null||turns<1)throw new IllegalStateException("players/turns must be posit
   Node p=tail.next;for(int i=0;i<turns;i++){if(i>0)System.out.print(" ");System.out.print(p.name);p=p.next;}Sys
 }
 public static void main(String[] args){CircularPlayers p=new CircularPlayers();p.addLast("Asha");p.addLast("Rav
}
```

Sample output

```
Asha Ravi Neha Asha Ravi Neha Asha
```

Complexity

addLast: O(1). printTurns(turns): O(turns).